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Experimental results of gradual porosity volumetric air receivers with wire meshes

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ABSTRACT

Central Receiver Systems with volumetric receiver technology has emerged as a scheme that could increase receiver and power block efficiencies. Among other choices, volumetric receivers made of metallic wire meshes are a promising technology that allows different designs and configurations. This paper details experimental work carried out on gradual porosity absorbers with the aim of promoting the basic principle of volumetric absorbers. In order to draw preliminary conclusions on the technology, a new lab-scale test bed of 1 kW was designed and used. As the absorber material, six commercial metallic wire mesh screens, of 310-alloy, with different geometrical and optical properties were selected with the constraint of finding pairs with similar porosities and different mesh properties. The thickness of each absorber was optimized in order to reduce the material needed to achieve complete absorption of the incident radiation. Moreover, pieces of two reference absorbers -TSA and SOLAIR-were tested. The concept shows potential for improving, or at least matching, the performance of the baseline absorbers (TSA and SOLAIR).

1. Introduction

Population growth, the increase in the consumption of energy producing larger CO_2 emissions and, the fossil fuel costs have strengthened interest in alternative, nonpetroleum-based, sources of energy. Among the various Solar Thermal Power Plant (STPP) technologies, Central Receiver Systems (CRSs) appear as a promising option. The first generation of commercial STPPs with CRS technology is based on technological developments based on more than two decades of research, using cavity or external tube receivers with saturated or superheated steam and molten salts schemes, respectively [1].

CRS with volumetric receiver technology using air as Heat Transfer Fluid (HTF) has not been marketed yet. However, it has emerged as a promising scheme that could increase receiver and power block efficiencies, both reducing radiative thermal losses and raising the temperature of the HTF as shown in several recent studies [2–12]. There are two basic power plant schemes in volumetric receiver applications which can include the aforementioned benefits [13–15]: open-loop receiver system with a Rankine cycle or closed-loop receiver system with Brayton cycle.

Volumetric receiver development dates back to the first prototype in 1983 [16]. The highly porous structure of volumetric receivers may be metal or ceramic. Ceramic is the most suitable option when temperatures above 800 °C are necessary [17,18]. On the other hand, metallic receivers are capable of producing a maximum air outlet temperatures of 800 °C with a faster response and lighter-weight structures [19].

Among all HTFs, atmospheric air has advantages in terms of availability, cost and environmental impact. The literature shows that it can be heated to temperatures around 1000 °C, generating steam at different working conditions in a heat recovery steam generator, which feeds a steam turbine paired with an electric generator [20,21].

For CRSs with volumetric receiver using atmospheric air as the HTF there are two baseline receivers: TSA for metallic and, SOLAIR for ceramic absorbers [18]. The TSA absorber tested at the Plataforma Solar de Almeria (PSA) was made up of Inconel 601 coiled knit-wire packs. The main result of the tests carried out in 1993 was the ability to achieve the design absorber efficiency of 85% at a receiver air outlet temperature of 700 °C, an average flux density of

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300 kW/m^2 and an air return ratio of 60%. The main conclusions from the tests of the 2.5 MW volumetric air receiver prototype were: 1) the system control was able to maintain constant air outlet temperatures, 2) the modular design would facilitate plant scale-up, 3) the absorbers were able to match the air flow distribution with the incident flux distribution, and 4) the process was flexible [22–24]. The SOLAIR absorber tested at the PSA was made of recrystallized silicon carbide (RSiC) with an open porosity of 49.5%. The receiver was a modular absorber assembled from 270 ceramic cups made of SiSiC. The results showed that the efficiencies varied from 70-to-75% at a receiver air outlet temperature of 750 °C for mean solar fluxes in the range of 370-to-520 kW/m^2 and an air return ratio of 50% [25]. As a result of the aforementioned results, in June 2006 it was decided to build a central receiver plant with volumetric receiver (with the SOLAIR technology) and thermal storage in Jülich, Germany. The aim of the project was to demonstrate this technology in a complete pre-commercial power plant for the first time [26–28].

In spite of these and other positive project results, most of the volumetric receivers tested during the last decades have not matched the predicted nominal conditions, working at lower efficiencies than expected. The project Sulzer 1 was the first volumetric receiver which due to its geometric design resulted in lower than expected efficiencies [29]. The project Sulzer 2 detected that the volumetric effect, which means that the absorber temperature at the irradiated side is lower than the outlet air temperature reducing the frontal thermal losses, was not totally satisfied [30]. The main innovation of the Catrec 1 project was its design with a small solid frontal area, in order to reduce the thermal losses [31], but despite this innovation the air outlet temperature measured were not as high as theoretically expected. With the second Catrec generation, one module of the prototype was irremediably burnt during the attempt to reach higher average air outlet temperatures than the achieved 460 °C [32]. Nevertheless, the technology has been given serious consideration for commercial implementation. The initial 10 MWe PS10 design was presented as an atmospheric air HTF circuit with open volumetric receiver technology [33]. Prior to the construction, the PS10 design was altered with a more-established technology with a water/steam generator. Moreover, as previously mentioned, the Jülich power tower plant is being running at a pre-commercial level in Germany. And recently, the AlSol research power plant has been proposed by a consortium for construction in Algeria [34].

But before commercial application of CRS with volumetric receiver is possible, there are some issues that need to be solved for the technology to be successful, i.e. the improvement of the volumetric effect and the solar absorption in the whole absorber, the adequate selection of the absorber porosity and geometrical characteristics to avoid the absorption just in the first millimeters, the optimization of the absorber thickness to use less material and produce lighter absorbers, further improvements on the performance and reliability of key components and materials durability under high solar fluxes.

As a consequence, this paper deals with some of these obstacles and describes the experimental work carried out for an innovative volumetric absorber concept, the "gradual decreasing porosity absorber", which had not been exhaustively studied previously. The gradual decreasing porosity volumetric absorber is a promising innovation that addresses the main thermal problems of the technology, as presented with computational fluid dynamic simulations in the literature [35,36]. The basic objectives of gradual porosity are to allow a deeper solar flux penetration, so that the maximum temperature is shifted further to the inside of the absorber, and reduce the frontal surface of the absorber which reduces the frontal thermal losses.

For that purpose, a new lab-scale test bed of 1 kWt coupled to a 4 kWe xenon short arc lamp was designed as a flexible and versatile approach from which preliminary conclusions could be drawn rapidly and effectively, with the target of improving the baseline volumetric absorbers: the TSA and SOLAIR absorbers.

To test the gradual decreasing porosity concept a metallic material was chosen due to its advantages in volumetric receivers and, ease of testing different geometries and configurations. A 310 alloy was selected as it has an ideal composition for working at high temperatures. Six commercial metallic plain-weave wire mesh screens were selected with different geometric characteristics, but with the constraint of finding pairs with similar porosities and different geometrical and optical properties. So the experimental tests analyze both the effect of the geometrical properties by means of the volumetric porosity and the specific surface area and, the optical properties by means of the extinction coefficient. Furthermore, with the 6 commercial plain-weave wire mesh screens selected, 26 volumetric absorbers −6 with single porosity, 12 with double porosity and 8 with triple porosity- and test pieces of the reference absorbers -TSA and SOLAIR-were tested and the results are presented. Moreover, the thickness of each one of the 26 volumetric absorbers was optimized in order to either reduce the material needed and to achieve complete absorption of the incident radiation in porous depth.

2. Plain-weave wire mesh screens properties

Structures made of metallic plain-weave wire mesh screens have been used in different applications including refrigeration, food processing, heating pipes, thermal insulation, solar heaters, regenerators and volumetric air receivers [37–42]. These structures are generally highly porous, with large specific surface area and with a highly tortuous flow path, and, when deployed in a CRS with volumetric receivers, they enhance the heat transfer between the porous matrix and the fluid circulating through it.

This section describes the geometrical (volumetric porosity, specific surface area, and hydraulic diameter) and optical properties (absorption coefficient, scattering coefficient, and extinction coefficient) of plain weave wire mesh screens. Both optical and geometrical properties influence the behavior of volumetric receivers; the geometrical properties, by means of the specific surface area, drive the convective heat transfer mechanisms [43], while the optical properties, by means of the extinction coefficient, drive the penetration depth of the incident radiation in the absorber [11].

2.1. Geometrical properties

The plain-weave wire mesh screen is the most commonly used weave type, in which each warp wire and shute wire pass over a wire and under an adjacent wire as depicted in Fig. 1 [19]. Fig. 1 summarizes the plan and edge views of a plain-weave screen unit cell where two geometric parameters that describe the mesh are the wire diameter, d and the mesh size, M. The wire filament pitch in the x and y directions are $M_x^{-1} = w_x + d_y$ and $M_y^{-1} = w_y + d_x$ respectively. In absence of crimping, the screen has a thickness $t = d_x + d_y$. When the mesh has a square screen: $d = d_x = d_y$; $M = M_x = M_y$ and $t = d_x + d_y = 2 \cdot d$.

The arrangement of plain-weave wire mesh screens is an important design parameter for volumetric absorbers. The literature shows two extremes for the arrangement of stacked plain-weave screens [44,45], the inline arrangement, Fig. 2 (a), where successive wire mesh screens are aligned and, the stagger arrangement, Fig. 2 (b), where successive wire mesh screens are offset in two directions by $0.5 \cdot M^{-1}$. The inline pattern has lower pressure drop and heat transfer coefficient and, higher porosity and

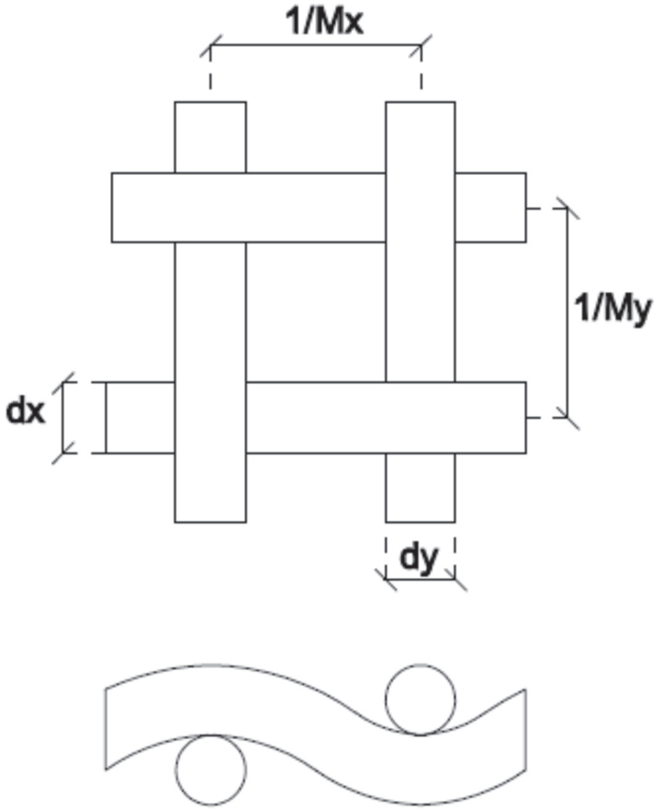

Fig. 1. Plain-weave unit cell.

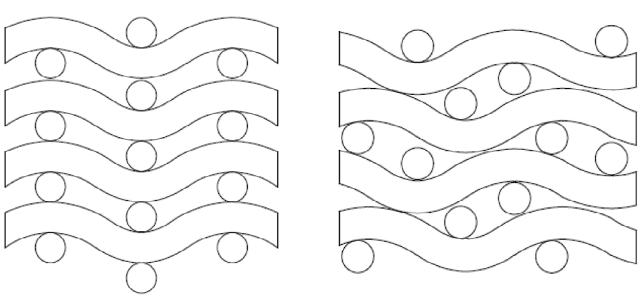

Fig. 2. Screen stacking configurations (a) Inline stacked; (b) Staggered stacked.

transmissivity, thus less frontal thermal losses, compared to the staggered pattern that has the opposite behavior [46].

An accurate estimation of the geometrical porous parameters undoubtedly helps to understand the convective heat transfer mechanisms inside those porous structures. This section presents the volumetric porosity, the specific surface area, and the mesh hydraulic diameter for both types of arrangements.

2.1.1. Volumetric porosity

The volumetric porosity, φ, of a porous structure is defined as the ratio of the void volume to the total volume of an average representative cell:

$$\varphi = 1 - \frac{V_s}{V_{tot}} \tag{1}$$

For a square plain-weave wire mesh screen the non-dimensional parameter C is defined as a function of the geometrical mesh parameters [19].

$$C = \left(\frac{M\cdot d}{2} + \frac{1}{2\cdot M\cdot d}\right)\cdot \arcsin\left(\frac{1}{\left(\frac{M\cdot d}{2}+\frac{1}{2\cdot M\cdot d}\right)}\right) \tag{2}$$

So the volumetric porosity of a square plain-weave wire mesh screen as a function of the geometric parameters is [19]:

$$\varphi = 1 - \frac{\pi\cdot M\cdot d\cdot C}{4\cdot C_f} \tag{3}$$

The constant C_f is the compactness factor and it is used to describe the thickness resulting from different types of packing pattern. When the wire mesh screens follow an inline pattern, the compactness factor equals one; when the wire mesh screens follows a staggered pattern, the compactness factor is less than one.

2.1.2. Specific surface area

Specific surface area, a_v, of a porous matrix is defined as the ratio of the total surface area to the total volume:

$$a_v = \frac{S_s}{V_{tot}} \tag{4}$$

Satisfying the criteria for square screens and using variable C (equation (2)), the specific surface area for a square plain-weave wire mesh screen is [19]:

$$a_v = \frac{\pi\cdot M\cdot C}{C_f} \tag{5}$$

2.1.3. Hydraulic diameter

Due to the extremely complex structure, it is important to select appropriate characteristic length for the porous media. Among the different length-scale parameters, the mesh hydraulic diameter is widely used [19,45,47], and it is also adopted in this study. The hydraulic diameter of a porous media is defined by Kaviany [48]:

$$d_h = \frac{4\cdot\varphi}{a_v} \tag{6}$$

2.2. Optical properties

Volumetric receivers work at high temperatures so the heat transfer by radiation plays an important role, especially in terms of frontal heat losses. The main optical properties considered in a porous media to model the radiative transfer equation are: the absorption coefficient, the scattering coefficient and, the extinction coefficient.

The radiative or optical properties of porous material have been widely investigated in the literature, either for ceramic porous materials [49–53] and metallic porous materials [54–56]. In this work, the optical properties were obtained using the geometrical optics approximation presented by Vafai [57]. The geometrical optics is an approximation that considers the main geometrical shape of the porous media for the evaluation of the optical properties. In the case of foams, the main shape is the sphere and the equations have been widely used and validated in the literature [11,35]. In the case of stacked wire meshes, the main shape of the porous media is a cylinder and the equations that hold are equations (7)–(9). These equations have been validated in the literature

[46] as well.

2.2.1. Absorption coefficient

The absorption coefficient is the property of the medium that describes the amount of absorption of thermal radiation per unit path length within the medium. It can be interpreted as the inverse of the mean free path that a photon will travel before being absorbed. The absorption coefficient is:

$$a = \frac{4}{\pi}\cdot\alpha\cdot\frac{(1-\varphi)}{d} \tag{7}$$

Where a is the absorption coefficient, α is the material absorptivity, φ is the porosity, and d is the wire diameter.

2.2.2. Scattering coefficient

The scattering coefficient is the property of a medium that describes the amount of scattering of thermal radiation per unit path length for propagation in the medium. It can be interpreted as the inverse of the mean free path that a photon will travel before undergoing scattering. The scattering coefficient is:

$$\sigma_s = \frac{4}{\pi}\cdot(2-\alpha)\cdot\frac{(1-\varphi)}{d} \tag{8}$$

Where σ_s is the scattering coefficient, α is the material absorptivity, φ is the porosity, and d is the wire diameter.

2.2.3. Extinction coefficient

The extinction coefficient is the sum of absorption coefficient and scattering coefficient and measure the capacity of a porous medium to absorb and scatter the radiation. The extinction coefficient is:

$$\beta = a + \sigma_s = \frac{8}{\pi}\cdot\frac{(1-\varphi)}{d} \tag{9}$$

Where β is the extinction coefficient, φ is the porosity, and d is the wire diameter.

All the properties presented depend on the physical characteristics of the mesh. As a result, Table 2 presents the main properties for inline and staggered stack arrangements.

As it is quite difficult to fulfill a perfect inline or staggered stack arrangement, particularly with those meshes with lower porosity and/or lower wire diameter (see Fig. 3 and Table 2), the volumetric absorbers were constructed following a random orientation of the meshes which is more similar to a staggered stack. As a result, the staggered stack parameters were adopted in this work.

3. Absorber materials

Volumetric absorbers are exposed to concentrated solar radiation, high temperature, daily temperature cycles and transients, etc. So, the construction materials requirements are:

- High solar absorption
- High heat transfer
- Low thermal gradients and thermal stresses
- High temperature resistance to any degradation

Taking into account the temperature operation and materials cost, alloy 310 was selected as it is a proper candidate to work at temperatures lower than 800 °C. Moreover, test pieces of the SOLAIR and TSA absorbers, made of re-SiC and alloy 601 respectively, were tested as benchmark absorbers.

Alloy 601 is a suitable material for applications that require thermal and corrosion resistance due to its high nickel and chrome content; moreover, it has the ability to form black oxides, increasing the absorptance. Alloy 310 is an austenitic stainless steel with chrome content similar to that of alloy 601, but better mechanical properties at high temperature. Table 1 presents the composition of both alloys. For temperatures higher than 1000 °C, the SiC is a suitable material, with a thermal conductivity of 78 W/m/K and a thermal expansion coefficient of 4.0 μm/(m·K).

For the investigation of graded porosity configurations, 6 alloy-310 commercial square plain-weave wire mesh screens were selected (Fig. 3) with a sample diameter of 50 mm. The meshes were chosen with the constraint of having pairs with similar porosities and different geometrical (see Section 2.1) and optical parameters (see Section 2.2), as presented in Table 2, for the two stacking configurations (inline and stagger). The wire diameter ranges from $0.13 < d < 1.00$ mm, and the mesh size varies from $0.20 < M < 3.03$ mm^{-1}. With these 6 meshes, 26 volumetric absorbers were constructed and tested: 6 with single porosity, 12 with double porosity and 8 with triple porosity following the pattern presented in Table 3.

The 26 samples were pre-oxidized before the test using the solar simulator. Therefore all the results obtained are likely to improve if the meshes are painted with a black coating. Comparing with ceramic foams, the meshes are easy to paint without blocking pores of the absorber as reported by Sandia [58]. Of course, this could be experimentally studied, which is however out of scope for the present work.

Table 1
Alloy 601 and 310 composition following the norms UNSN06601 and UNS S31000 respectively.

Alloy	Ni	Cr	Fe	Other
601	58–63	21–25	Balance	Al 1.0–1.7
310	19–22	24–26	Balance	Si 1.5/Mn 2.0

4. Experimental methodology

This section presents the experimental methodology carried out. Firstly, the methodology use to set the absorber thickness of each sample is presented. Then, the measurement of the incident heat flux and the test bed are described. Finally, the thermal evaluation methodology and the quasi-steady states selection are shown.

4.1. Absorber thickness

The thickness of the volumetric absorbers is a key-design parameter as the literature shows that solar absorption usually happens in the first millimeters [18]. For this reason, the transmissivity (τ) of each mesh ($mesh_i$) of each absorber ($absorber_j$) was measured and the thickness of each sample reached its maximum value when the transmissivity of the absorber was less than the measurement error or reached a null value.

$$\tau = \frac{\left(\int_{-y}^{y}\int_{-x}^{x}\varphi_{inc,\ z}(x,y)\cdot dx\cdot dy\right)_{mesh_i/absorber_j}}{\left(\int_{-y}^{y}\int_{-x}^{x}\varphi_{inc,\ z}(x,y)\cdot dx\cdot dy\right)_{no-absorber}} \tag{10}$$

The adopted criterion was that each porosity, of each composition, should absorb equal amount of radiation: 1) when the

Table 2
Commercial AISI 310 square plain-weave wire mesh characteristics.

Mesh type				A	B	C	D	E	F
Mesh properties		d	mm	1.00	0.70	0.50	0.16	0.63	0.13
		M	mm^{-1}	0.20	0.31	0.53	1.79	0.61	3.03
Geometrical properties	Inline stack	φ	%	84	82	78	76	67	66
		a_v	m^{-1}	645	1013	1729	5910	2114	10476
	Stagger stack	φ	%	70	68	62	62	48	47
		a_v	m^{-1}	1194	1849	3044	9552	3322	16330
Optical properties	Inline stack	β	m^{-1}	411	645	1101	3763	1346	6669
	Stagger stack	β	m^{-1}	760	1177	1938	6081	2115	10396

Table 3
Number of meshes for the 26 absorber samples tested.

Absorber samples	N° meshes		
	Porosity 1	Porosity 2	Porosity 3
A	10	–	–
B	9	–	–
C	8	–	–
D	8	–	–
E	5	–	–
F	6	–	–
AC	2	7	–
AD	2	8	–
AE	2	5	–
AF	2	6	–
BC	2	8	–
BD	2	7	–
BE	2	5	–
BF	2	5	–
CE	2	5	–
CF	2	6	–
DE	2	5	–
DF	2	5	–
ACE	2	1	4
ACF	2	1	5
ADE	2	1	6
ADF	2	1	6
BCE	2	1	4
BCF	2	1	5
BDE	2	1	5
BDF	2	1	5

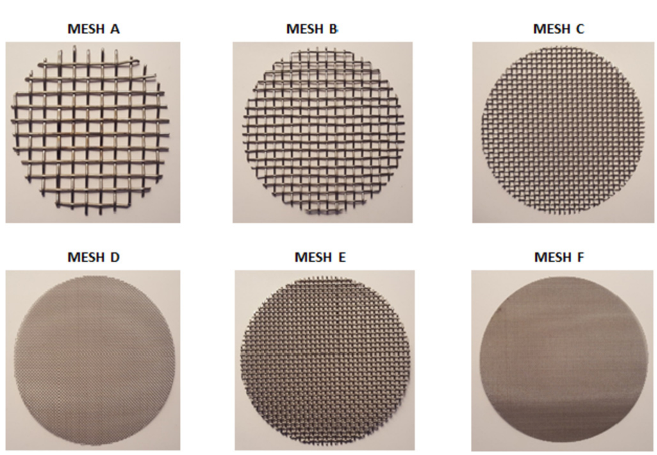

Fig. 3. Commercial square plain-weave wire mesh screens for the analysis of new absorber designs.

absorbers were of single porosity, a 100 ± 1% had to be absorbed in the whole structure, 2) when the absorbers were of double porosity, a transmissivity of 50 ± 1% was allowed for each type of porosity and, 3) when the absorbers were of triple porosity, a transmissivity of 33 ± 1% was allowed for each type of porosity.

Two Gardon radiometers were used to measure the concentrated radiation flux. The measurement error of the single device is about 3%, but when the measurements of two radiometers were compared, the maximum deviation was lower than 1% without considering systematic errors. Moreover, all the measurements were repeated at least three times and the composition with the highest number of meshes were adopted.

Fig. 4 shows the transmissivity as a function of the number of meshes of three compositions: single (A), double (AC) and triple porosity (ACE). The single porosity absorber with type-A mesh needed ten meshes to reduce all of the incident radiation in the porous volume. For the double porosity absorber AC, two type-A meshes and seven type-C meshes were needed. And finally for the triple porosity absorber ACE, two type-A meshes, 1 type-C mesh and 4 type-E meshes were required for the complete absorption of the incident radiation. With this criterion, the transmissivity behind each sample was null or within the error of the radiometer (1%).

4.1.1. Absorber samples

As described previously, 6310-alloy commercial screens (Fig. 3) were used to construct and test 26 volumetric absorbers. The meshes the absorbers are touching with no space between them. Table 3 shows the number of screens needed to construct the 26 volumetric absorbers: firstly the 6 absorbers with single porosity, secondly the 12 absorbers with double porosity and finally the 8 with triple porosity. With the arrangement presented in Table 3 the radiation behind each sample is null or within the measurement error of the Gardon radiometer (1%) and the thermal equilibrium is

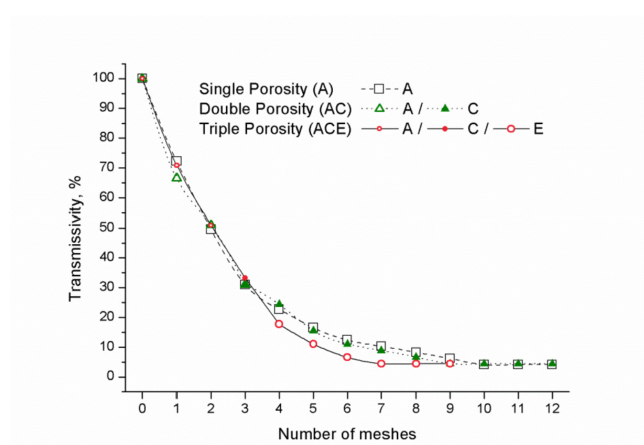

Fig. 4. Transmissivity measurements for a single, double and triple porosity absorber as function of the number of stacked meshes.

ensured as presented by Avila-Marin [59].

4.2. Incident heat flux

The incident heat flux is supplied by a 4 kWe Xenon-arc lamp that consists of a point-shaped cathode and a rounded-shaped anode contained in a spherical quartz bulb. This lamp is coupled to an optical reflector to produce an intense beam of concentrated radiant energy in the focus [60]. The emitted radiation is confined to a rim angle of 60° and the elliptical reflector has a truncation diameter of 0.3 m.

The heat flux was measured daily using any of the two available Gardon radiometers interchangeably [61] with an error of less than ±1%. The radiometer is placed in a positioner that sweep a square plane of 60 mm × 60 mm and record 81 measurements that are interpolated to obtain the average incident heat flux and incident power over the sample.

The incident heat flux on the frontal surface of the samples ranges from 420 to 400 kW/m^2, and the incident power ranges from 830 to 780 W for a sample diameter of 50 mm.

4.3. Facility description

Previous authors have written that more than thirty volumetric receivers with nominal power between 200 and 3000 kWt, had been tested, mostly at the PSA [18]. Testing of this type of absorber carries with it both high economic cost and a great deal of labor and, because of this, a test bed had been developed to test 1 kWt absorbers, which vary in size from 40 to 60 mm diameter, as a flexible and versatile approach from which preliminary conclusions could be drawn rapidly and effectively.

The test bed is composed of four sub-systems as presented in Fig. 5 and Fig. 6 [9]:

- A 4 kWe solar simulator made up of a xenon lamp and a parabolic concentrator. Further details about the xenon lamp have been described by Hirsch et al. [60], Petrasch et al. [61] and Gallo et al. [62].
- A receiver sub-system where the volumetric absorbers of 40, 50 or 60 mm are placed. It is equipped with 24 K-type thermocouples distributed in six sections, 1 T-type thermocouple, 2 PT100 surface sensors and an infrared camera.

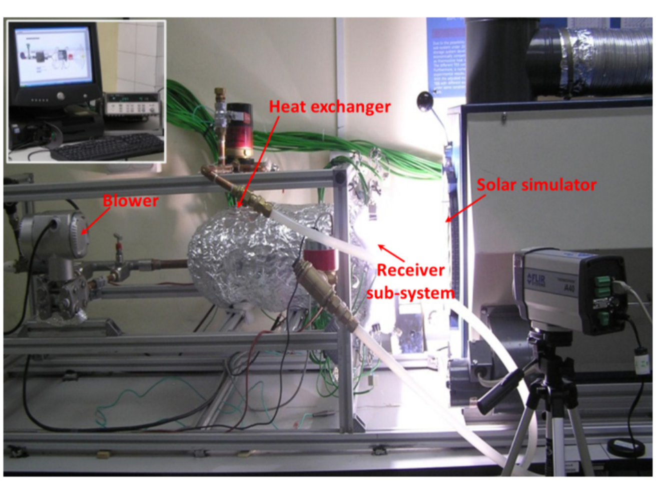

Fig. 5. Experimental devices for the thermal evaluation of open volumetric receivers with a 4-kW xenon arc lamp.

- A helical air-water heat exchanger sub-system equipped with 4 PT100 sensors, 2 PT100 surface sensors, 1 T-type thermocouple, a water mass flow-rate measurement, and a pump.
- An extraction sub-system with 1 PT100 sensor, an air mass flow-rate measurement, and a blower.

The blower forces the ambient air to flow through the volumetric absorber as it heats up. Then, the air is the energy carrier fluid which circulates counter current in a water heat exchanger in order to reduce the air temperature to protect the blower and to get an indirect measurement of the receiver efficiency.

The test of each sample took between 6 and 8 h. During this time the irradiation was kept constant. To achieve various air outlet temperatures, the air-volume-flow was varied four times. At the start-up, the highest value (7 m^3/h) was set to avoid thermal shocks. After each step the mass flow was kept constant between 90 and 120 min to reach thermal equilibrium. The reason for this long period is that despite the air achieved the steady state quite quickly, the water of the heat exchanger needed more time to get steady conditions. Once the steady state was reached, the temperatures and the radiation data were taken to characterize this steady state period. After 4 steps the lowest mass flow (4 m^3/h) was set with the corresponding "hottest" operating point. Then, each sample is tested at least two more times to demonstrate a general repeatability of the efficiency measurement.

4.4. Thermal evaluation

The thermal evaluation of the test carried out could be performed in two different ways as shown in the literature [63–65]:

- Analysis of the receiver performance: The main assumption made for the thermal evaluation of the receiver is to consider that the power gained by the air in the receiver is transmitted to the water in the heat exchanger, in steady state conditions. This methodology requires the estimation of the air temperature at the receiver outlet (or the heat exchanger inlet), as performed in the Sulzer test-bed installed in PSA [9,30,32,66,67]. Then, the receiver efficiency is obtained.
- Analysis of the absorber performance: The main assumption made for the thermal evaluation of the absorber is to consider that the power gained by the air, just in the absorber, is the sum of the air power at the receiver outlet and the thermal losses throughout the receiver. This methodology requires:

o Firstly, the estimation of the air temperature at the receiver outlet/heat exchanger inlet, as shown in "Analysis of the receiver performance" [9].
o Secondly, the estimation of the air temperature at the absorber outlet, as performed for the TSA receiver tested in PSA [65].
o Lastly, the computation of the absorber efficiency.

Both methodologies leads into the same trends and conclusions, therefore, the second methodology is adopted in this paper, as proposed by Haeger et at. in their evaluation of the TSA receiver, for comparison with numerical simulations of the absorber in future works. Moreover, at lab scale, it is easily possible to evaluate the thermal losses of the receiver for a better understanding of the absorber performance. Then, two variables were used to compare the absorbers: the absorber efficiency and the absorber mean air outlet temperature, which are the variables most commonly used.

4.4.1. Receiver and absorber efficiency

The receiver efficiency is defined as the ratio between the receiver power output and the receiver power input. To get more

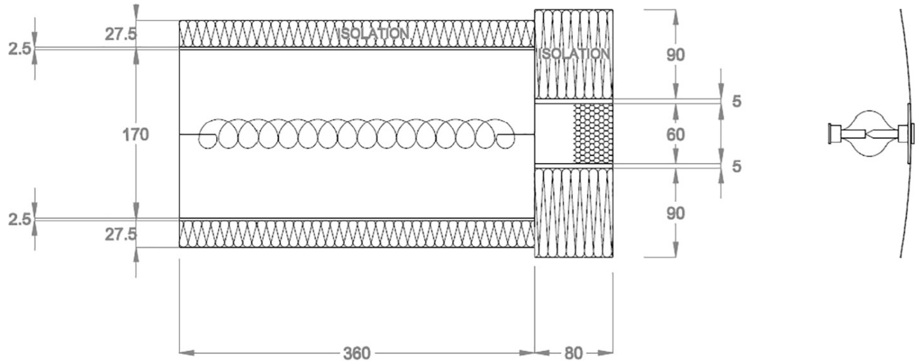

Fig. 6. Sketch of the receiver and the heat exchanger with dimensions in mm.

information about the receiver performance an analysis of the absorber has to be made. The absorber efficiency is defined as the ratio between the power transferred by the absorber to the air and the incident power in the frontal surface of the receiver [65]:

$$\eta_{abs} = \frac{\dot{m}\cdot\int_{T_{amb}}^{\overline{T}_{air-abs-out}} c_{p_{air}}\cdot dT_{air}}{\overline{P}_{inc,\ z=cte}} \tag{11}$$

Thermal efficiency is a function that depends on the technological design of the absorber, the experimental conditions (orientation, distance to the focal plane, etc.) and the operational working conditions (output temperature regimen, input power flux, etc.). Thus, the absorber thermal efficiency should be evaluated under different scenarios. At lab scale most of the working and operational conditions are controlled, so the methodology consisted in the variation of the volumetric flow rate under similar incident flux.

Incident flux was measured by a calibrated water-cooled calorimeter, and the evaluation of the average incident flux and the average incident power were calculated with equations (12) and (13).

$$\overline{\varphi}_{inc,\ z=cte} = \int_{-y}^{y}\int_{-x}^{x} \varphi_{inc,\ z=cte}(x,y)\cdot dx\cdot dy \tag{12}$$

$$\overline{P}_{inc,\ z=cte} = \overline{\varphi}_{inc,\ z=cte}\cdot A_{abs} \tag{13}$$

Where $\overline{\varphi}_{inc,\ z=cte}$ is the mean incident flux measured in a constant plane, $\overline{P}_{inc,\ z=cte}$ is the mean incident power and, A_{abs} is the frontal surface.

4.4.2. Absorber air outlet temperature

Although several thermocouples were installed in the test-bed, it is not possible to know exactly the mean air outlet temperature behind the absorber. So it is necessary to estimate it according to the following methodology, similar to that presented by Haeger [65]:

- Calculation of the power gained by the water in the heat exchanger:P_{w-HE}.
- Calculation of the thermal losses in the heat exchanger: TL_{HE}.
- Calculation of the air power at the heat exchanger inlet/receiver outlet: $P_{air-rec} = P_{w-HE} + TL_{HE}$.
- Calculation of the mean air outlet temperature from the receiver: $\overline{T}_{air-rec-out}$.
- Calculation of the thermal losses in the receiver: TL_{rec}.
- Calculation of the air power at the absorber outlet: $P_{air-abs} = P_{air-rec} + TL_{rec}$.
- Calculation of the mean air outlet temperature from the absorber: $\overline{T}_{air-abs-out}$.

The power gained by the water in the heat exchanger and the thermal losses in both the heat exchanger and the receiver are calculated with the available thermocouples, while the mean air outlet temperature in both the receiver and the absorber are calculated using energy balances as shown in next sections.

4.4.2.1. Power gained by the water in the heat exchanger. The power gained by the water in the heat exchanger is calculated as a function of the water temperature difference between the heat exchanger outlet and inlet and the water mass flow rate through the heat exchanger. The variables are evaluated in steady state conditions and the power is computed with equation (14).

$$P_{w-HE} = \dot{m}_w\cdot\int_{T_{w_{out-HE}}}^{T_{w_{in-HE}}} c_{p_w}(T_w)\cdot dT_w \tag{14}$$

Where P_{w-HE} is the power gained by the water in the heat exchanger, $\dot{m}_w$ is the mass flow rate, c_{p_w} is the heat capacity of the water at constant pressure, $T_{w_{in-HE}}$ is the heat exchanger inlet water temperature and, $T_{w_{out-HE}}$ is the heat exchanger outlet water temperature.

4.4.2.2. Power gained by the air in the receiver. The power gained by the air in the receiver is the sum of the power gain by the water in the heat exchanger and the thermal losses throughout the heat exchanger.

$$P_{air-rec} = P_{w-HE} + TL_{HE} \tag{15}$$

Where $P_{air-rec}$ is the power gained by the air in the receiver and, TL_{HE} is the thermal loss in the heat exchanger.

The two PT100 surface sensors installed in the heat exchanger (the first in the upper-front surface and the second in lower-back surface) confirm that the alumina blanket isolation installed in the heat exchanger (Fig. 6), with a composition of 50% Al_2O_3 and 50% SiO_2 and a density of 0.096 g/cm^3, was sufficient to consider the thermal losses through the heat exchanger negligible. When the thermal losses are null, as in the cases presented in this paper, the heat exchanger is adiabatic ($TL_{HE} = 0$) and the power gained by the water is the same to the power gained by the air in the receiver [9].

$$P_{air-rec} = \dot{m}_w \cdot \int_{T_{w_{out-HE}}}^{T_{w_{in-HE}}} c_{p_w}(T_w) \cdot dT_w \tag{16}$$

4.4.2.3. Calculation of the mean air outlet temperature from the receiver. Once the power gained by the air in the receiver is computed with equation (16), the air temperature at the receiver outlet or the heat exchanger inlet ($\overline{T}_{air-rec-out}$), is easily obtainable using iteration in the following energy balance.

$$P_{w-HE} = P_{air-rec} = \dot{m}_w \cdot \int_{T_{w_{out-HE}}}^{T_{w_{in-HE}}} c_{p_w}(T_w) \cdot dT_w = \dot{m}_{air} \cdot \int_{T_{amb}}^{\overline{T}_{air-rec-out}} c_{p_{air}}(T) \cdot dT\left(T_{amb}; \overline{T}_{air-rec-out}\right) \tag{17}$$

For the application of this energy balance, the temperature of the air at the exit of the heat exchanger is used, which is nearly the ambient temperature.

4.4.2.4. Power gained by the air in the absorber. The power gained by the air in the absorber is the sum of the power of the air at the receiver outlet (or the power gained by the water in the heat exchanger) and the thermal losses throughout the receiver, computed with equation (18). The thermal losses in the receiver are produced by an air temperature reduction between the absorber outlet and the receiver outlet due to insufficient alumina blanket isolation.

$$P_{air-abs} = P_{air-rec} + TL_{rec} \tag{18}$$

Where $P_{air-abs}$ is the power gained by the air in the absorber and, TL_{rec} is the thermal loss in the receiver.

The thermal losses of the receiver are useful when computing the power gained by the air only through the contribution of the absorber (Fig. 7) and for the numerical analysis of the different absorbers configurations in future works.

Two PT100 surface sensors where installed in the receiver outer surface (the first in the upper-front surface and the second in lower-back surface) and measure the surface temperature in the receiver system. The average temperature where used for the estimation of the thermal losses by convection ($TL_{conv,rec}$) and radiation ($TL_{rad,rec}$).

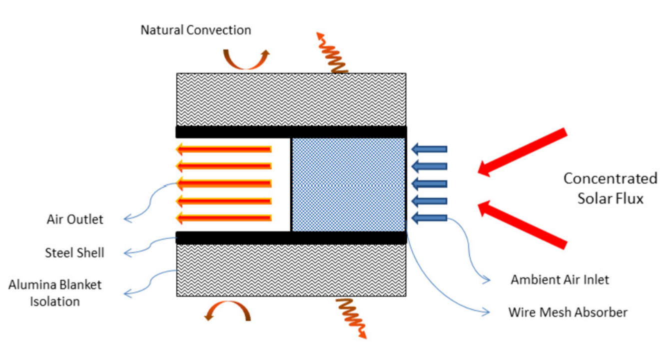

Fig. 7. Absorber placed in the receiver with the main heat transfer mechanisms and components.

$$TL_{rec} = TL_{conv,rec} + TL_{rad,rec} \tag{19}$$

4.4.2.4.1. Thermal losses by convection. The thermal loss by convection through the receiver is evaluated with the following expression:

$$TL_{conv,rec} = \pi \cdot d_{blanket\_rec\_ext} \cdot L_{rec} \cdot h_{ext} \cdot \left(\overline{T}_{rec,ext} - T_{amb}\right) \tag{20}$$

As the tests are performed at lab scale, there is no wind and the convection heat transfer from the receiver to the environment is due to natural convection. For this case, the correlation developed by Churchill and Chu is used to estimate the Nusselt number [68] as the literature shows that it can be used for similar purposes as the support bracket of a parabolic trough [69].

$$Nu_{blanket} = \left\{0.60 + \frac{0.387 \cdot Ra_{blanket}^{1/6}}{\left[1 + (0.559/Pr)^{9/16}\right]^{8/27}}\right\}^2 \tag{21}$$

$$Ra_{blanket} = \frac{g \cdot \beta \cdot \left(\overline{T}_{blanket} - T_{amb}\right) \cdot d_{blanket\_ext}^3}{\alpha \cdot \upsilon} \tag{22}$$

Where $Nu_{blanket}$ is the Nusselt number, $Ra_{blanket}$ is the Rayleigh number for the air based on the blanket outer diameter, $Pr_{blanket}$ is the Prandtl number, g is the gravitational constant, β is the volumetric thermal expansion coefficient, $d_{blanket\_ext}$ is the outer diameter of the alumina blanket used as insulation for the receiver, $T_{blanket}$ is the surface temperature of the alumina blanket, T_{amb} is the ambient temperature, α is the thermal diffusivity for the air and υ is the kinematic viscosity for the air.

The parameter β is evaluated at the temperature of the fluid for gases and all other the properties are evaluated at the average temperature (T_m):

$$T_m = \frac{\overline{T}_{blanket} + T_{amb}}{2}$$

This correlation is valid for $10^{-5} < Ra \le 10^{12}$ and $0 < Pr < \infty$.

4.4.2.4.2. Thermal losses by radiation. The thermal loss by radiation through the receiver is caused by a temperature difference between the blanket and the environment. To approximate this, the envelope is assumed to be a small convex gray object that emits to the atmosphere (blackbody). The net radiation transfer between the blanket and the environment becomes [68]:

$$TL_{rad,rec} = \pi \cdot d_{blanket\_rec\_ext} \cdot L_{rec} \cdot \varepsilon_{blanket} \cdot \sigma \cdot \left(\overline{T}_{rec,ext}^4 - T_{amb}^4\right) \tag{23}$$

Where $\varepsilon_{blanket}$ is the emissivity of the blanket, σ is the Stefan-Boltzmann constant number.

4.4.2.5. Calculation of the mean air outlet temperature from the absorber. Once the power gained by the air in the absorber is known with equation (18), the mean air outlet temperature from the absorber ($\overline{T}_{air-abs-out}$) can be calculated using iteration in the following energy balance:

$$P_{air-abs} = \dot{m}_{air} \cdot \int_{T_{amb}}^{\overline{T}_{air-abs-out}} c_{p_{air}}(T) \cdot dT\left(T_{amb}; \overline{T}_{air-abs-out}\right) \tag{24}$$

4.4.3. Selection of quasi-steady states

After analyzing the results, it is important to select the quasi-steady states for a clear comparison of the variables. This criterion is particularly useful for solar radiation and solar simulators due to their fluctuations.

In this work the criterion adopted said that for a certain sampling interval (d), each measure at any time (t) is compared to the five previous measurements, allowing the following variation:

- $(F(t) - F(t-d))/F(t) < 1\,\%$
- $(F(t) - F(t-2d))/F(t) < 2\,\%$
- $(F(t) - F(t-3d))/F(t) < 3\,\%$
- $(F(t) - F(t-4d))/F(t) < 4\,\%$
- $(F(t) - F(t-5d))/F(t) < 5\,\%$

Where F(t) is the value of any variable at the time t and F (*t*-d) is the value of the same variable in the previous sampling interval (d). With these assumptions, variable changes are negligible compared with the absolute value of their correspondent variables during a relevant period of time.

Any variable that fulfills the previous requirements is in quasi-steady state at time t and can be used for comparison against the variables obtained for other absorbers in similar working conditions. The air temperatures measure with the 24 K-type thermocouples inside the receiver sub-systems and the water temperatures measure with 4 PT100 sensors inside the heat exchanger are used to set up the quasi-steady states.

5. Results and discussion

The experimental work carried out consisted of testing 26 absorbers −6 with single porosity, 12 with double porosity and 8 with triple porosity− and 2 baseline absorbers −TSA and SOLAIR− at four different volumetric flow rates and similar incident flux. The number of screens used for each one of the 26 absorbers is presented in Table 3. All the test data are presented in a scatter plot together with the tendency lines.

The following sub-sections presents the results obtained for absorbers with single, double and triple porosity divided by groups and finally, a comparison of the best combinations against the baseline absorbers (TSA and SOLAIR) is shown.

5.1. Absorbers with single porosity

The comparison of six single porosity absorbers (using the commercial meshes presented in Table 2 and Fig. 3 and with the number of screens presented in Table 3) is shown in this section. Moreover, as an example that both methodologies presented in Section 4.4 leads into the same trends and conclusions, this section presents both results: absorber efficiency (solid lines and filled scatter plot) and receiver efficiency (dashed lines and hollow symbols) as function of the absorber air outlet temperature and receiver air outlet temperature respectively.

As a result, Fig. 8 depicts the absorber and the receiver efficiency as function of the absorber mean air outlet temperature and the receiver mean air outlet temperature for the 6 single porosity absorbers. The analysis of the results shows that:

- The absorber with single porosity and mesh type C is the one with the highest efficiencies.
- The absorber with single porosity and mesh type F is the one that behaves the worst, having the lowest temperatures.

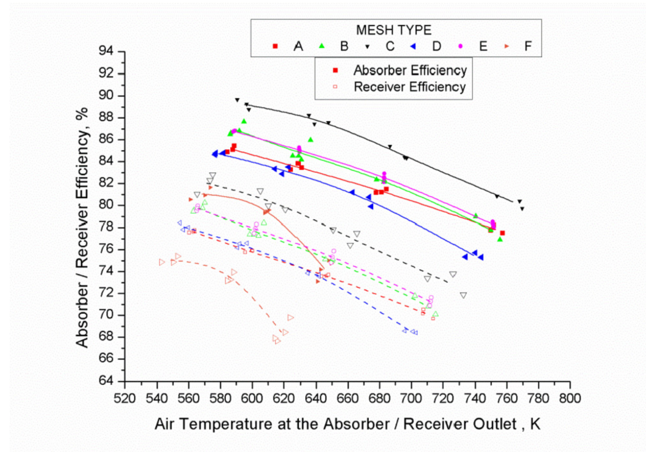

Fig. 8. Results of the absorber and receiver efficiency for the 6 absorbers with single porosity.

- The other absorbers can be divided in two sub-groups, one for the absorbers with meshes type B and E with a higher efficiency compared to absorbers with meshes A and D, the second sub-group. Among then, the one with mesh type D has the lowest efficiency.

From the previous results, it can be concluded:

- A very high extinction coefficient (related to the solar transmissivity through the depth of the absorber), likes meshes type F and D, produces worse thermal behavior despite having a large specific surface area (related to a better heat transfer coefficient). The reason are: 1) those meshes with very large extinction coefficient and specific surface area are the meshes with the smallest wire diameter that apparently are similar to bi-dimensional receiver as depicted in Fig. 3 and, 2) those meshes prevent the solar penetration in the depth of the absorber, producing larger frontal thermal losses. Moreover, the porosity plays an important role, as the high the porosity is the better the performance of the absorber is.
- The lower the specific surface area and extinction coefficient, independently of the porosity, the poorer the performance of the absorber as happens with absorber type A. The low efficiency is explained because of the low specific surface area that produces and inefficient transmission of the heat flux from the solid matrix to the air despite the better solar absorption. However, the behavior is better than the extremely high properties as absorbers type F and D.
- Furthermore, intermediate properties for the specific surface area and extinction coefficient together with a high porosity provides the best results as happens with absorber type C. Absorber type E has similar intermediate properties than absorber type C but the poorer porosity makes the receiver to perform worse, as previously shown for absorber type F compared to absorber type D.
- Finally, it can be concluded that the best performance is related to the higher possible porosity with intermediate properties in terms of specific surface area and extinction coefficient. As a low extinction coefficient is linked to a low specific surface area that increases the solar absorption in the depth but avoids an efficient heat transfer, while a high extinction coefficient together with a high specific surface area reduces the solar absorption in

the depth of the absorber despite having an efficient heat transfer.

5.2. Absorbers with double porosity

This section presents the experimental results for 12 absorbers with double porosity. The analysis is divided in three different plots depending on the base-mesh (first porosity) of the absorbers: 4 absorbers with mesh base type A, 4 absorbers with mesh base type B and, 2 absorbers with mesh base type C and 2 absorbers with mesh base type D that are presented together.

Each base mesh is combined with those meshes that have lower volumetric porosity (see Section 4.1.1). So, the base-meshes type A and B were used to construct four double porosity absorbers each (AC, AD, AE, AF and BC, BD, BE, BF), while with the base-meshes type C and D just two absorbers of each type were built (CE, CF and DE, DF). The number of meshes used for each absorber is presented in Table 3.

5.2.1. Double porosity with base mesh type A

This section compares four absorbers with double porosity using mesh type A as base (or as first porosity). Fig. 9 depicts the absorber efficiency as function of the mean air outlet temperature for the absorbers AC, AD, AE and AF. The analysis of the results shows that:

- All the double porosity absorbers with base mesh A improve on the performance of the absorber with single porosity with mesh type A.
- The greatest improvement is produced by configuration AF, which produces the highest efficiencies.
- The remainders of the absorber combinations have decreasing efficiencies depending on the specific surface area of the mesh located behind the base mesh A.

Once the first porosity is selected (mesh type A), the best choice is to have as the second porosity of the absorber the one with the highest specific surface area possible, as happens with configuration AF.

Moreover, the configuration AF combines the most extreme meshes of this study (Table 2). On one hand, base mesh type A with the lowest specific surface area and extinction coefficient, allowing the highest solar penetrability toward the inside of the receiver, while the second porosity (located in the back part of the absorber), mesh type F, present the opposite properties and, is the responsible of the improvement of the heat transfer between the mesh and the air of the radiation transmitted by type A mesh.

Moreover, for these combinations of absorbers (AC, AD, AE, AF), the specific surface area of the mesh located at the back determine the overall performance of the configuration. The higher the specific surface area of the mesh located at the back is the higher thermal efficiency of the absorber.

5.2.2. Double porosity with base mesh type B

Turning to the comparison of the four absorbers using mesh type B as a base, Fig. 10 depicts the absorber efficiency as function of the mean air outlet temperature for the absorbers BC, BD, BE and BF. Similar results, to those obtained with the double porosity absorbers with base mesh type A, can be observed.

The BF double porosity absorber is the one that clearly improves the efficiency of the single porosity absorber with type B mesh the most. Then, the BD configuration is the second with the better performance with a small improvement. The remainders of the configurations match (BC) or performs slightly worse (BE) than the single porosity absorber with mesh type B.

It is seen, with a different base mesh to the one presented in the previous section, that once the first porosity is selected, the results are better when the second porosity (located at the rear part of the absorber) has as low porosity, high specific surface area and high extinction coefficient as possible.

Again, the best configuration is the one that combines the most extreme meshes of this sub-section, the configuration BF. The same approach presented previously is valid for the configuration in terms of the influence of the geometrical and optical properties over the thermal behavior.

The double porosity absorbers with mesh type B do not improve the single porosity absorber B as much as the double porosity absorbers with mesh type A do. The main reason is that base mesh B has a higher extinction coefficient than mesh A, allowing a lower solar penetrability in the depth of the absorber, diluting the effect of the second porosity.

The configuration BF, the best double porosity with type B base mesh, does not improve on the results of the AF configuration, the best double porosity with type A base mesh. Despite the mesh B offers as high porosity as mesh type A and better specific surface area (related to a better heat transfer coefficient), the poorer performance is explained by the higher extinction coefficient of mesh

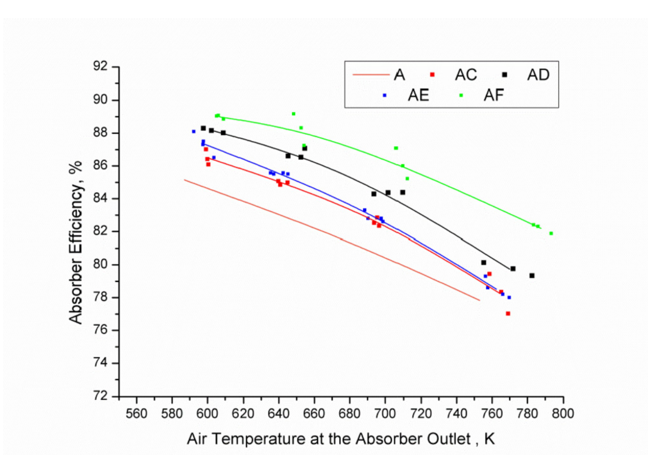

Fig. 9. Results of the absorber efficiency for 4 absorbers with double porosity with mesh A as first porosity.

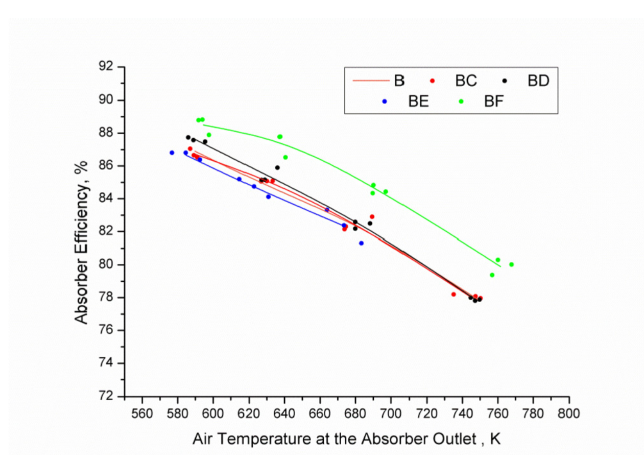

Fig. 10. Results of the absorber efficiency for 4 absorbers with double porosity with mesh B as base.

type B compared to mesh type A (nearly 1.5 times higher) which produces a lower solar absorption of the incident flux.

5.2.3. Double porosity with base mesh type C and base mesh type D

In this section the results of the two absorbers constructed with mesh type C as base (CE and CF) and the two absorbers constructed with mesh type D as base (DE and DF) are presented together, as their performance is quite similar. Fig. 11 depicts its efficiencies as function of the mean air outlet temperature.

The double porosity absorbers with type C and D base meshes show slight and minimal improvement over the single porosity designs respectively. Even so, slightly positive tendencies in the absorbers with rear meshes of type F are seen. This is due to the fact that the base geometries are hard to improve on because of their geometrical and optical properties. Type C mesh is the best single porosity absorber; consequently the incorporation of type F meshes produces a slight improvement. The absorber with type D mesh, with very large extinction coefficient, gives rise to high thermal losses coming out of the absorber and the effect of incorporating meshes behind it can be considered equally insignificant.

5.3. Absorbers with triple porosity

This section presents the experimental results for the 8 absorbers with triple porosity. The analysis is divided in two subsections depending on the base-mesh of the absorbers: 4 absorbers with mesh base type A and, 4 absorbers with mesh base type B.

The triple porosity designs are based on the double porosity designs AC and AD, to which are added meshes of lower porosity according to the following plan: high porosity/average porosity/low porosity. The number of screens used for each composition is presented in Table 3.

5.3.1. Triple porosity with base mesh type A

In this section a comparison of four absorbers with triple porosity using mesh type A as first porosity is presented. Fig. 12 shows the absorber efficiency as function of the mean air outlet temperature for the triple porosity absorbers ACE, ACF, ADE and, ADF. Moreover, this figure includes the results of the double porosity absorbers AC and AD, as they are the absorbers used to construct the triple porosity compositions. The analyses of the results show that:

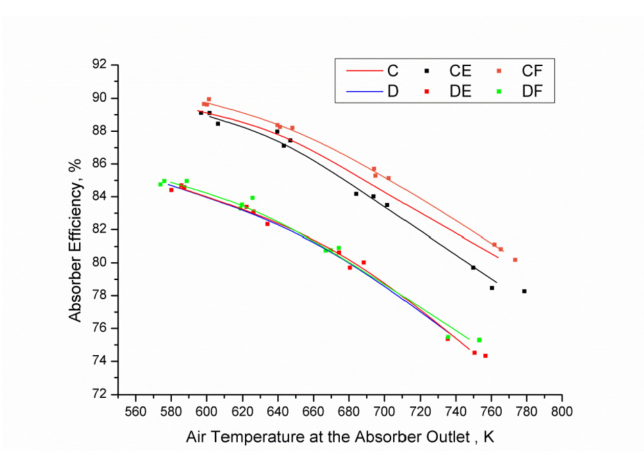

Fig. 11. Results of the absorber efficiency for 2 absorbers with double porosity with mesh C as base and for 2 absorbers with double porosity with mesh D.

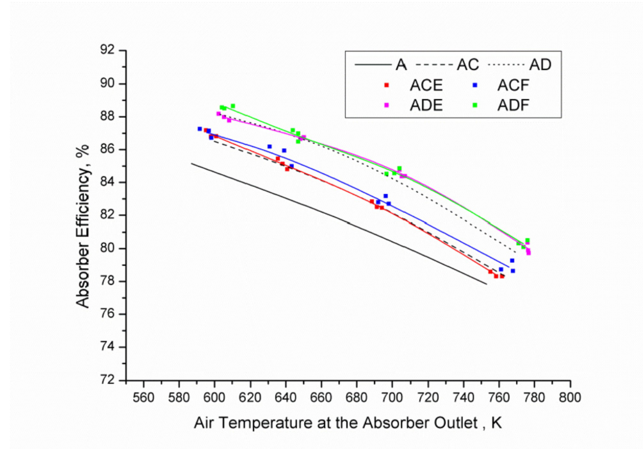

Fig. 12. Results of the absorber efficiency for 4 absorbers with triple porosity with mesh A as base.

- All the triple porosity absorbers improved on the behavior of the single porosity absorber with mesh type A.
- The triple porosity designs based on the AC configuration match or slightly improve on the AC base absorber, while those triple porosity designs based on the AD configuration improve the AD base absorber slightly.
- Among all the triple porosity designs, the absorbers that show the best results are the ADF and ADE configurations, in that order.
- The two pairs of triple porosity configurations, ACE-ACF and ADE-ADF have very similar performance both between the members of the pairs and with their respective double porosity configurations of AC and AD.

From the previous results, it can be concluded that:

- The last porosity of the triple porosity absorbers has only a slight benefit on performance. This benefit is higher when the second porosity has as high specific surface area as possible, as more heat is transferred toward the third porosity.
- The configurations that perform better are those that have a porosity that decreases with the thickness of the volumetric absorber. Moreover, its behavior is improved when the specific surface area has a value that increases with the thickness, especially in the average porosity region.
- The efficiency of the absorber improves when the last mesh of each composition is of type F. Even so, the improvement that is produced with respect to the AC and AD double porosity absorbers is very slight.

5.3.2. Triple porosity with base mesh type B

Turning to the comparison of the four absorbers using mesh type B as a base, Fig. 13 depict the absorber efficiency as function of the mean air outlet temperature for the triple porosity absorbers BCE, BCF, BDE and BDF. Moreover, this figure includes the results of the double porosity absorbers BC and BD, as they are the absorbers used to construct the triple porosity compositions. Results similar to those absorbers with base mesh type A can be observed:

- All the triple porosity absorbers improved on the behavior of the single porosity absorber with mesh type B.
- The triple porosity designs based on the BC configuration slightly improve on the BC base absorber, while those triple

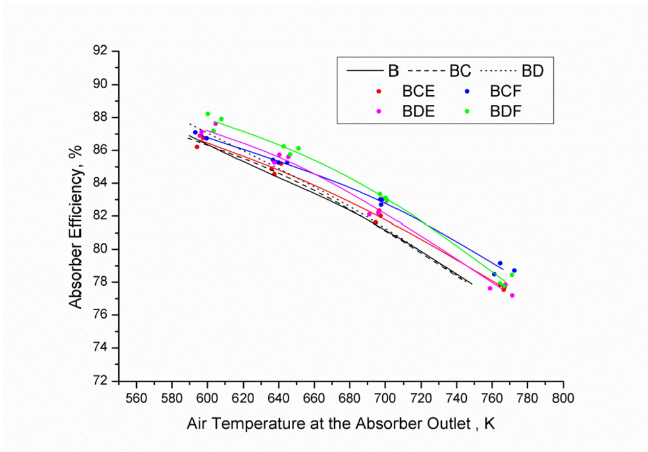

Fig. 13. Results of the absorber efficiency for 4 absorbers with triple porosity with mesh B as base.

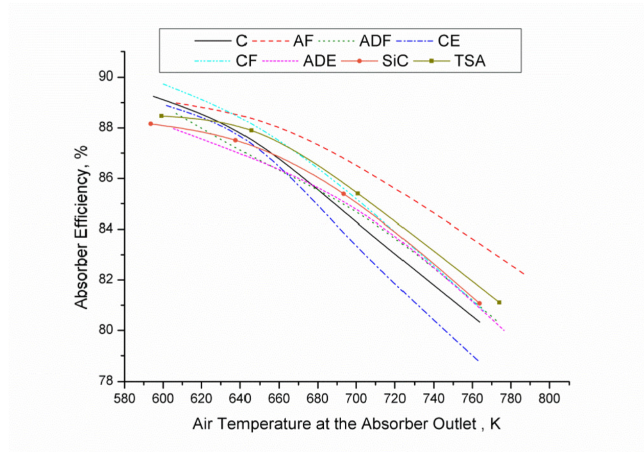

Fig. 14. Results of the absorber efficiency for the best six absorbers tested and the baseline absorbers (TSA and SOLAIR).

porosity designs based on the BD configuration improve in between 1 and 2% the BD base absorber.

- From the two pairs of triple porosity configurations, BCE-BCF and BDE-BDF, the best performance is achieved when the last mesh of each composition is of type F.
- Among all the triple porosity designs, the absorbers that show the best results are the BDF, BCF and BDE configurations, in that order.
- Due to the geometrical and optical characteristics of the first porosity mesh, presented in Section 2, the results are closer but are still clear.

The same conclusions reach regarding the triple porosity absorbers with mesh type A as base apply to the equivalent configurations based on mesh type B. The best configurations are those that have a porosity that decreases with the thickness of the volumetric, improving their behavior when the specific surface area has a value that increases with the thickness, especially in the second porosity region (average porosity).

Moreover, the best triple porosity configurations obtained with mesh B (BDF, BCF and BDE) does not improve on the results from the ADE and ADF configurations, the best triple porosity configurations with mesh A as base. Its poorer performance is due to the location of the meshes for the second porosity, which are nearer the frontal surface (increasing the thermal losses) with type B configuration than the meshes of type A configuration.

5.4. Comparison against the baseline absorbers

This section compares the performance of the baseline absorbers, TSA for volumetric absorbers with metallic material and SOLAIR for volumetric absorbers with ceramic material, with the best configurations tested.

Fig. 14 show the absorber efficiency as function of the mean air outlet temperature for the six best absorbers tested (C; CE; CF; AF; ADF; ADE) together with the absorbers reported in the literature to be the baseline, the TSA and SOLAIR. As the curves show:

- Only the AF absorber offers greater efficiency than the reference absorbers for all the working conditions.
- Absorbers C, CE and CF only offer greater efficiency than the reference absorbers at high flow rates.
- Last, the ADF absorber has lower efficiency than the reference absorbers except at the highest flow rate tested while the ADE performs worse than the reference absorber at all working conditions.

This comparison shows that there are different designs and configurations that perform similarly or even better than the reference absorbers with a simpler design, lower weight and reduced thickness. Moreover, the graded porosity concept, with configuration AF, is able to improve the performance of both baseline absorbers, demonstrating the potential of the concept. Capuano et al. [3] presented similar conclusions to those presented in this work for an innovative design with graded porosity that minimizes the radiative thermal losses and, allows a good penetration of the solar radiation.

6. Conclusions

The present work has started a line of research on volumetric absorbers with gradual porosity at the experimental level with the development of 26 prototypes that have been exhaustively tested. Until now there has been no experimental work in the literature similar to that presented here analyzing that concept.

In addition, the concept of graded porosity has been demonstrated to have potential, in the absence of an optimization process, which can improve, or at least match, the efficiency of the baseline volumetric absorbers at a much lower cost, using less material and producing lighter absorbers.

The geometrical and optical properties of the screens used have a strong influence on the thermal behavior of the absorber. On the one hand, the geometrical properties, by means of the specific surface area, drive the convective heat transfer mechanisms, while, on the other hand, the optical properties, by means of the extinction coefficient, drive the absorption of the incident radiation throughout the absorber.

Although some of the observed changes of efficiency are inside the confidence interval (especially in terms of power input, with a maximum deviation lower than 1%), the relative variations of the efficiency gives information about the trends. Moreover, the measurements use the same procedure and the tests are repeatable.

The analysis of the results shows that,

- For single porosity absorbers, the best performance is related to the higher possible porosity with intermediate properties in terms of specific surface area and extinction coefficient, as:

- o A low extinction coefficient (linked to a low specific surface area) increases the solar absorption in the depth but avoids an efficient heat transfer and,
- o A high extinction coefficient (linked to a high specific surface area) reduces the solar absorption in the depth of the absorber despite having an efficient heat transfer.

- For double porosity absorbers, it is concluded that:

- o On the one hand, if the first porosity is selected, the best behavior is achieved when the second porosity of the absorber has the highest specific surface area possible.
- o On the other hand, if the first porosity is not selected, the performance of the absorber is maximized when the first porosity has as high porosity and low extinction coefficient as possible, together with a second porosity with the highest specific surface area possible.

- For triple porosity absorbers, it is concluded that:

- o They are not able to improve the best double porosity configurations, due to the extreme properties of the double porosities absorbers tested compared with the triple ones.
- o The best performance is achieved when the porosity decreases with the thickness and the specific surface area increases. Especially in the intermediate porosity region.

This research demonstrates that even in absence of an optimization process, there are several absorbers that perform similarly or ever better to the baseline receivers presented in the literature, TSA and SOLAIR. More specifically, 6 configurations of the 26 absorbers tested have demonstrated similar results, having 1 with single porosity (C), 3 with double porosity (AF, CD, CE) and 2 with triple porosity (ADF, ADE).

Finally, according to the experimental results, the thermal behavior could be further increase using a double porosity absorber that uses for the first porosity a mesh similar to type A, but with the highest possible porosity, together with a mesh similar to type F, but with the highest possible specific surface area, for the second porosity.

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Nomenclature

A	Frontal surface area (m^2)
C_f	Compactness factor $(-)$
C	Characteristic mesh constant $(-)$
d	Diameter (m)
d	Sampling interval (s)
d_h	Hydraulic diameter (m)
g	Gravity constant (m/s^2)
L	Length (m)
M	Mesh size (1/m)
$\dot{m}$	Mass flow rate (kg/s)
P	Power (W)
S_s	Wire surface area (m^3)
T	Temperature (K/°C)
$\overline{T}$	Temperature (K/°C)
T_m	Average temperature (K/°C)
t	Thickness (m)
T	Time (s)
V	Volume (m^3)
W	Distance between two wires (m)
a_v	Specific surface area (1/m)
a	Absorption coefficient (1/m)
σ_s	Scattering coefficient (1/m)
α	Absorptivity $(-)$
α	Thermal diffusivity (m^2/s)
β	Extinction coefficient (1/m)
β	Volumetric thermal expansion coefficient $(1/K)$
ε	Emissivity $(-)$
c_p	Heat capacity $(J/(kg\cdot K))$
σ	Stefan-Boltzmann constant number $(W/(m^2\cdot K^4))$
η	Efficiency (%)
τ	Transmissivity $(-)$
υ	Kinematic viscosity (m^2/s)
θ	Intersection angle (°)
φ	Porosity $(-)$
φ_{inc}	Incident flux (W/m^2)

Abbreviations

inc	Incident
R	Recrystallized
SiC	Silicon carbide
Ni	Nickel
Cr	Chrome
Fe	Iron
Al	Aluminium
Si	Silicon
Mn	Manganese
N	North
E	East
S	South
W	West
NE	Northeast
SE	Southeast
SW	Southwest
NW	Northwest
Nu	Nusselt number
Ra	Rayleigh number
Pr	Prandtl number
TC	Thermocouple
TL	Thermal losses
HE	Heat exchanger
kWt	Thermal power
kWe	Electrical power

Subscripts

abs	Absorber
in	Inlet
conv	Convection
ext	External
out	Outlet
rad	Radiation
rec	Receiver
s	Solid
tot	Total
w	Water

x X direction
y Y direction
z Z direction

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